

Lierda NT21-KCN A Series Hardware Design Manual

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Document Revision History

Document version	Change Date	Reviser	Reviewer	Change content
Rev1.0	25-01-21	HKJ	SLY、YMX	Official version
Rev1.1	25-02-11	HKJ	SLY、YMX	Explanation on why adding a TVS diode in power supply design is indispensable;
Rev1.2	25-02-21	HKJ	SLY、YMX	Modify the size of the carrier rubber wheel; Modify the description of the shutdown section;
Rev1.3	25-02-24	HKJ	SLY、YMX	Modify the module silkscreen;
Rev1.4	25-11-21	HKJ		Specify frequency band information and weight; Modify the RF matching series position;
Rev1.5	25-12-08	HKJ		Correct the wrong pin number;

Safety Instructions

Users are responsible for complying with the relevant regulations of other countries regarding wireless communication modules and devices, as well as specific regulations related to the usage environment. By following the safety principles below, personal safety can be ensured, and it helps protect the product and work environment from potential damage. Our company does not assume any liability for losses resulting from customers' failure to comply with these regulations.



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Please turn off mobile terminal devices before boarding. The wireless functions of mobile terminals are prohibited from being activated on the aircraft to prevent interference with the aircraft's communication systems. Ignoring this instruction may compromise flight safety and even violate the law.



When in hospitals or healthcare facilities, pay attention to any restrictions on the use of mobile terminal devices. RF interference can cause medical equipment to malfunction, so it may be necessary to turn off mobile terminal devices.



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Please keep mobile terminal devices away from flammable gases. When you are near gas stations, oil depots, chemical plants, or explosive operation sites, please turn off mobile terminal devices. Operating electronic devices in any location with potential explosion hazards poses safety risks.

Applicable Module Selection

Serial Number	Module Model	Feature symbol	Supported frequency bands	Dimensions (mm)	Module Introduction
1	NT21-KCN	NT21KCNA0MNNA	Band3/8/ 34/38/39/40/41	13.45×10.48 ×2.2	Sleep Mode: VDD_EXT power down

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1 Introduction

This document defines the application specifications for the NT21-KCN A series Cat.1 wireless communication module. It describes its frequency band functions, key features, hardware interfaces, operating modes, electrical characteristics, mechanical specifications, as well as packaging and storage. It can help users quickly grasp the application methods of this Cat.1 module and carry out product design quickly and flexibly.



2 Product Overview

The NT21-KCN A series module is a globally leading Cat.1 wireless communication module, compliant with the 3GPP R14 Cat.1bis standard. It supports bandwidths of 1.4/3/5/10/15/20MHz and features a compact size, low power consumption, and strong anti-interference capability. The module is suitable for various common IoT application scenarios, such as:

- ◆ Intelligent Meter Reading
- ◆ Intelligent Parking
- ◆ Smart City
- ◆ Intelligent Security
- ◆ Asset Tracking
- ◆ Smart Home Appliances
- ◆ Agriculture and environmental monitoring, etc.

2.1 Frequency Bands and Functions

The wireless network functions supported by the NT21-KCN A series modules are as follows:

Table 2-1 Module Support for Wireless Network Functions

Network Function	Explanation
LTE	B3/B8
LTE	B34/B38/B39/B40/B41

Precautions

LTE-TDD B41 only supports 140M (2535-2675MHz).

2.2 Key Features

The key features of the NT21-KCN A series module are as follows:

Table 2-2 Description of Key Module Characteristics

Parameters	Explanation
Module Packaging	LGA
Module size	13.45mm×10.48mm×2.2mm (L×W×H)
Module Weight	Approximately 0.8g
Operating voltage	◆ Power supply voltage: 3.3 to 4.5V, typical value: 3.8V ◆ Extended power supply voltage (1): 3.1~4.5V
Operating temperature	◆ Normal operating temperature range (2): -35 to 75°C ◆ Extended operating temperature range (3): -40 to 85°C
Storage temperature	Storage temperature range(4): -40~90°C
VDD_EXT feature	Sleep mode: VDD_EXT power down
Ultra-low power consumption	Sleep mode: 6uA@3.8V
Transmit power	23dBm±2.7dB (Max)
Antenna interface	50Ω characteristic impedance
USIM	Supported USIM card types: Class B (3.0 V) and Class C (1.8 V)
UART	Main serial port: ◆ Used for AT command transmission and data transfer, the supported baud rate defaults to 115200bps. ◆ Used for firmware upgrades, the supported baud rate defaults to 921600bps.
	Debugging serial port: ◆ Used for software debugging ◆ The default baud rate of the serial port is 3Mbps.
	Auxiliary serial port: ◆ User-defined functions
USB	◆ Compatible with USB 2.0 (slave mode only), data transfer rate up to 480 Mbps ◆ Used for AT command transmission, data transfer, software debugging, and

	firmware upgrades
Communication Interface Characteristics	Supports 3GPP Rel.13/14 Cat.1 radio communication interface and protocols
Network Protocol Features	Supports protocols such as TCP/UDP/HTTP(S)/SSL/MQTT(S)/FOTA/PPP/RNDIS/FTP/ECM.
Data Transmission Characteristics	◆ LTE-FDD: Maximum downlink speed 10Mbps, maximum uplink speed 5Mbps ◆ LTE-TDD: Maximum downlink rate 8.96 Mbps, maximum uplink rate 3.1 Mbps
Firmware Upgrade	◆ Main Serial Port Upgrade ◆ USB Interface Upgrade ◆ DFOTA Upgrade
RoHS	All components comply with the EU RoHS standard.

Remarks

- (1) For a detailed description of the extended voltage, see section 6.2.
- (2)(3)(4) For specific descriptions regarding temperature, see section 6.6.

2.3 Functional block diagram

The NT21-KCN A series modules feature a wide range of peripheral interfaces and RF functions. The functional block diagram is as follows:

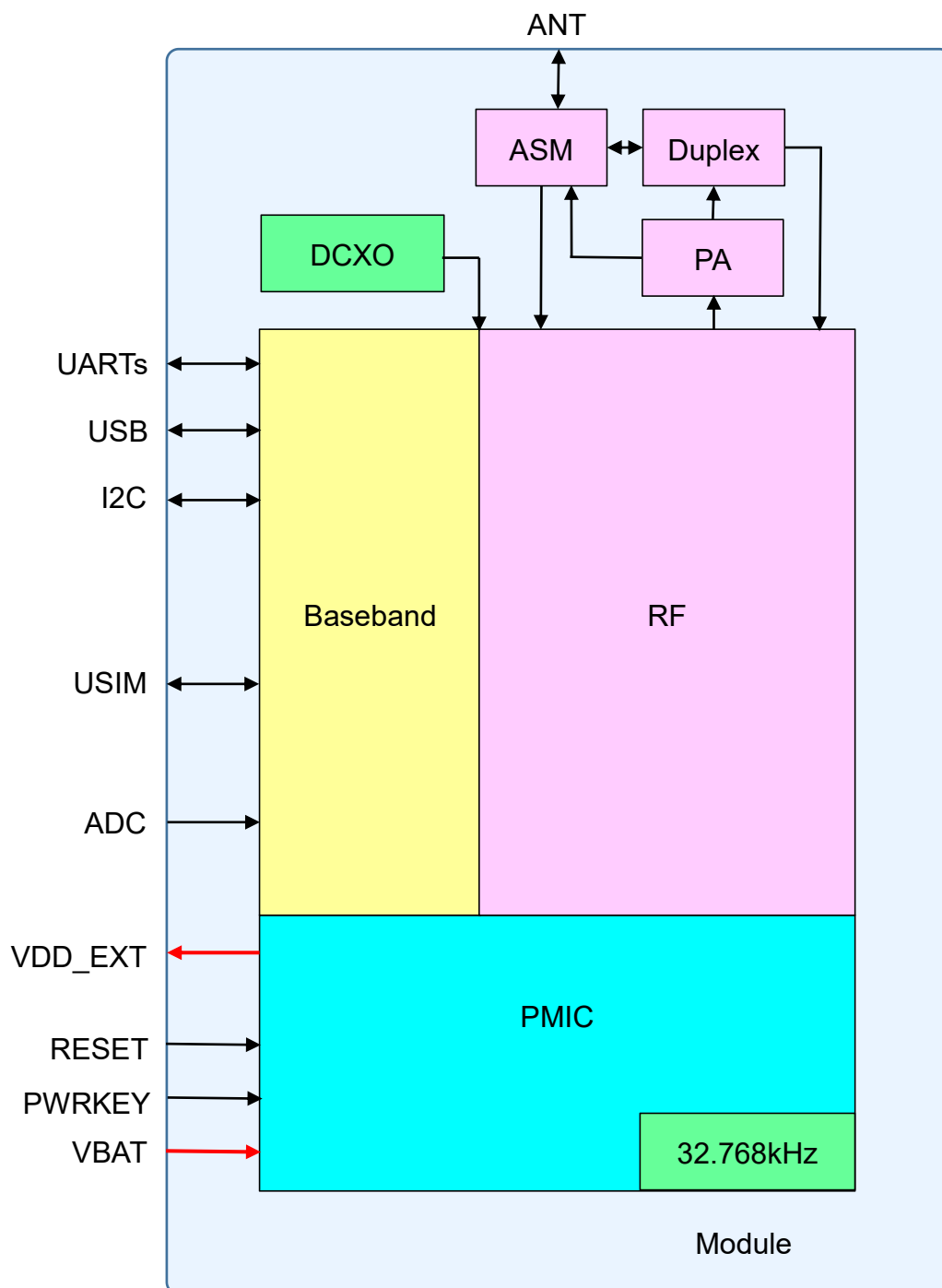


Figure 2.1 Module Function Block Diagram

2.4 Pin distribution diagram

The NT21-KCN A series module has a total of 50 LGA pins. The module pin distribution diagram is as follows:



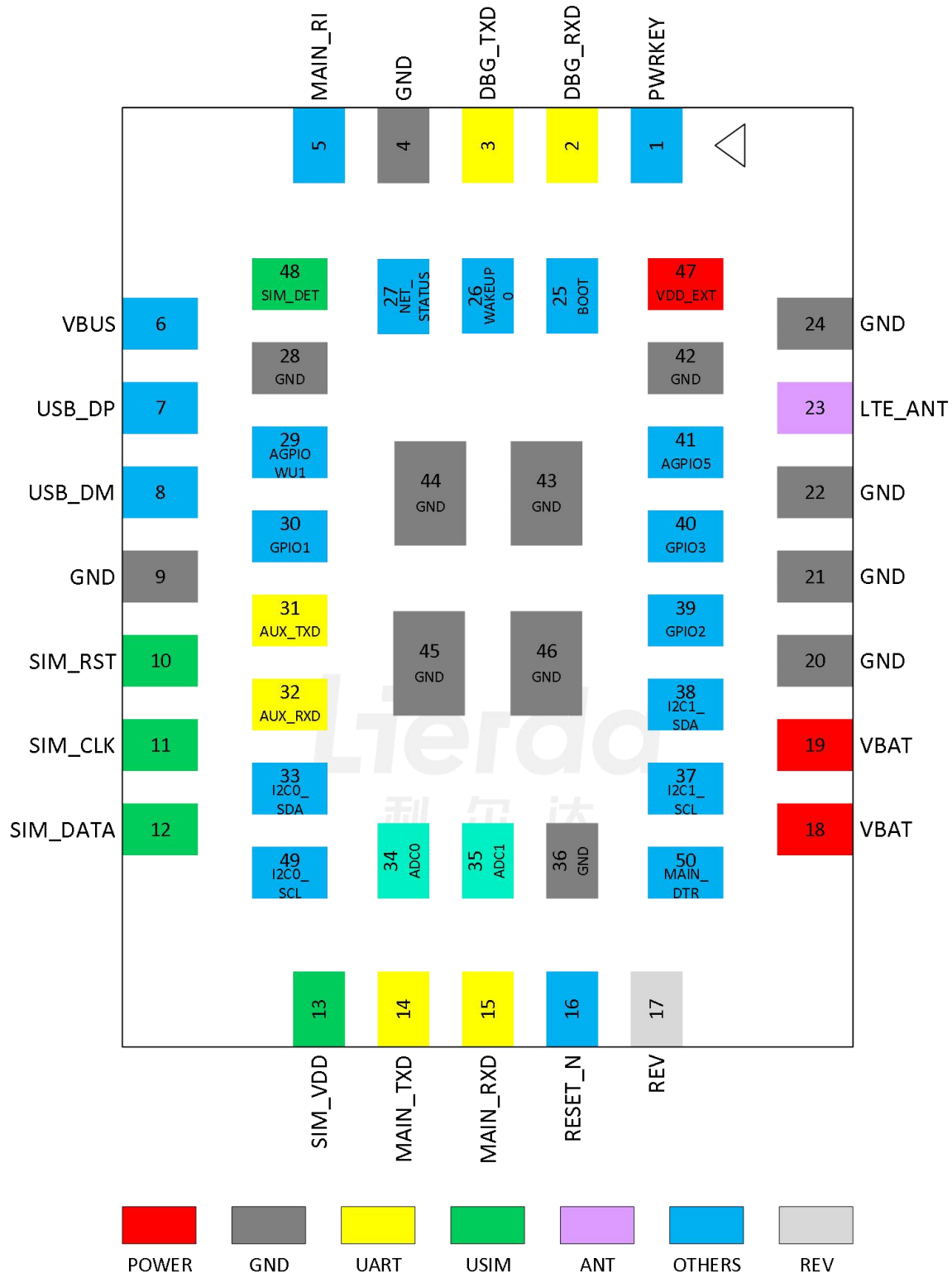


Figure 2.2 Module Pin Assignment Diagram

2.5 Pin Description Table

For better understanding of the application, the following table provides the type

definitions of the I/O parameters:

Table 2-4 Type Definition Description of I/O Parameters

I/O Parameter Types	Explanation
DI	Digital input
DO	Digital output
DIO	Digital Input and Output
AI	Simulated input
AO	Simulation output
AIO	Simulated input and output
PI	Power input
PO	Power output
G	Grounding

Table 2-5 Module Power Domain Characteristics Description

Power domain parameters Type	DC Characteristics	Explanation	Power supply target
VBAT	Vmax=4.5V Vmin=3.3V Vnorm=3.8V	Module power input, it is recommended to use a 3.8V/1.2A power supply.	Module
VDD_EXT	VILmax=0.2×VDD_EXT VIHmin=0.7×VDD_EXT VOLmax=0.15×VDD_EXT VOHmin=0.8×VDD_EXT	Default 1.8V; Different hardware versions: In the sleep mode VDD_EXT power-off version, the normal driving capability is 120mA, with no output in sleep mode. In sleep mode, the VDD_EXT non-power-down version has a driving capability of 3 to 5mA.	VDD_EXT

VO_LDOIO	VOLmax=0.2×VO_LDOIO VOHmin=0.7×VO_LDOIO VILmax=0.15×VO_LDOIO VIHmin=0.8×VO_LDOIO	Default 1.8V; In sleep mode, VO_LDOIO powers down.	UART GPIO
VDD18AON	VILmax=0.2×VDD18AON VIHmin=0.7×VDD18AON VOLmax=0.15×VDD18AON VOHmin=0.8×VDD18AON	No power loss in sleep mode	WAKEUP RESET
LDO_AONIO	VILmax=0.2×LDO_AONIO VIHmin=0.7×LDO_AONIO VOLmax=0.15×LDO_AONIO VOHmin=0.8×LDO_AONIO	No power loss in sleep mode	AGPIO
USIM_VDD	Vnorm=1.8/3.0V VOLmax=0.15×USIM_VDD VOHmin=0.8×USIM_VDD VILmax=0.2×USIM_VDD VIHmin=0.7×USIM_VDD	Dedicated power supply for SIM cards, supporting 1.8/3.0V cards.	USIM

Table 2-6 Module Pin Description

1. Power					
Pin number	Pin name	I/O	Description	DC Characteristics	Remarks
18,19	VBAT	PI	Power supply	Vnorm=3.8V	The external power supply needs to provide a current carrying capacity of more than 1.2A.
47	VDD_EXT	PO	Output power	Vnorm=1.8V Iomax=120mA	Can be used for external circuit pull-up or reference level. There are two hardware versions in sleep mode: power-down and non-power-down, with power-down as the default; leave floating if not used, and it is

					recommended to reserve a test point.
4,9,20,21,22,24, 28,36,42,43~46	GND	G	Grounding	-	-

2. Power On/Off and Reset

Pin number	Pin name	I/O	Description	DC characteristics	Remarks
1	PWRKEY	DI	Module power on/off pins	$V_{ILmax}=0.45V$	Internal pull-up, active low
16	RESET_N	DI	Module reset pin	$V_{ILmax}=0.3V$	Internal pull-up, active low

3、USB

Pin number	Pin name	I/O	Description	DC Characteristics	Remarks
6	VBUS	DI	External USB 5V input	$V_{max}=5.25V$ $V_{min}=3.0V$ $V_{nom}=5.0V$	In sleep mode, USB insertion will wake up the module.
7	USB_DP	DIO	USB differential data (+)	-	If not used, leave floating
8	USB_DM	DIO	USB differential data (-)	-	

4. Serial Port UART

Pin number	Pin name	I/O	Description	DC characteristics	Remarks
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15	MAIN_RXD	DI	Main serial port receiving data	VO_LDOIO	
14	MAIN_TXD	DO	Main serial port sends data	VO_LDOIO	
50	MAIN_DTR	DI	Main serial port data terminal ready	VDD18AON	Wake-up pin can be used in sleep mode
5	MAIN_RI	DO	Main serial port outputs ring indication	VDD18AON	If not used, leave floating
2	DBG_RXD	DI	Debugging serial port data reception	VO_LDOIO	It is recommended to reserve test points.
3	DBG_TXD	DO	Debugging serial port data transmission	VO_LDOIO	It is recommended to reserve test points.
32	AUX_RXD*	DI	Assist with serial port data reception	VO_LDOIO	Not supported yet
31	AUX_TXD*	DO	Assist with serial port data transmission	VO_LDOIO	Not supported yet

5. External USIM Card Interface

Pin number	Pin name	I/O	Description	DC Characteristics	Remarks
10	SIM_RST	DIO	SIM card reset line	USIM_VDD	
11	SIM_CLK	DIO	SIM card clock line	USIM_VDD	
12	SIM_DATA	DIO	SIM card data cable	USIM_VDD	
13	SIM_VDD	PO	SIM card power supply	1.8/3.0V	
48	SIM_DET	DI	SIM card status signal detection	VDD18AON	Active high

6. Antenna Interface RF

Pin	Pin name	I/O	Description	DC	Remarks
-----	----------	-----	-------------	----	---------

number				characteristics	
23	LTE_ANT	AIO	RF antenna interface	50Ω characteristic impedance	

7. I2C Interface

Pin number	Pin name	I/O	Description	DC characteristics	Remarks
33	I2C0_SDA	DIO	I2C Serial Data	VO_LDOIO	If not used, leave floating
49	I2C0_SCL	DO	I2C serial clock	VO_LDOIO	If not used, leave floating
38	I2C1_SDA	DIO	I2C serial data	VO_LDOIO	If not used, leave floating
37	I2C1_SCL	DO	I2C Serial Clock	VO_LDOIO	If not used, leave floating

8. ADC Interface

Pin number	Pin name	I/O	Description	DC Characteristics	Remarks
34	ADC0	AI	General ADC Interface	Internal direct connection: Input voltage range: 0~1.05V	If not used, leave floating
35	ADC1	AI	General ADC Interface	Internal division: Input voltage range: 0~3.3V Default 1.05V The internal voltage division can be enabled through the software.	If not used, leave floating

9. Status Indicator

Pin number	Pin name	I/O	Description	DC Characteristics	Remarks
27	NET_STATUS	DO	Network Status Indicator	VDD18AON	If not used, leave floating

10. Other interfaces

Pin number	Pin name	I/O	Description	DC Characteristics	Remarks
25	BOOT	DI	Emergency download mode control. Before powering on the module, pull up to VO_LDOIO; after the module starts, it will enter emergency download mode.	VO_LDOIO	The module has a default weak pull-down internally and can be used as a GPIO after power-on; if not used, it can be left floating.
26	WAKEUP0	DI	Wake-up pin	VDD18AON	If not used, leave floating
29	AGPIOWU1	DIO	General-purpose GPIO, can be maintained in sleep mode	LDO_AONIO	If not used, leave floating
30	GPIO1	DIO	General GPIO	VO_LDOIO	If not used, leave floating
39	GPIO2	DIO	General GPIO	VO_LDOIO	If not used, leave floating
40	GPIO3	DIO	General GPIO	VO_LDOIO	If not used, leave floating
41	AGPIO5	DIO	General GPIO	LDO_AONIO	If not used, leave floating

11. Reserved interface					
Pin number	Pin Name	I/O	Description	DC characteristics	Remarks
17	REV	-	-	-	Undefined pin, it is recommended to leave it floating.

2.6 Evaluation Kit

Lierda can provide complete evaluation and development kits, including module adapter boards, Cat.1 & NB-IoT module universal development boards, etc. Please feel free to contact us for inquiries.

3Work Characteristics

3.1 Working mode

The NT21-KCN A series modules have multiple operating modes. The table below briefly describes several common operating modes of the modules:

Table 3-1 Work Modes

Mode	Status description
Normal working mode	Data transmission: Network connection is normal. Module power consumption depends on the network environment and data transmission rate.
	Idle state: The software is running normally. The module is connected to the network but has no data interaction with the network.
Minimum Function Mode	AT+CFUN=0 can set the module to minimum functionality mode, where both the RF and USIM card do not operate.
Sleep mode	The module's power consumption will be reduced to an extremely low level, but the module can still receive paging, short messages, and TCP/UDP data.
Shutdown mode	VBAT power supply is continuous, software stops working

Remarks

For AT+CFUN related commands, please refer to the AT command data manual.

3.2 Sleep mode

The module can enter sleep mode through AT commands. In sleep mode, the module can reduce power consumption to an extremely low level, while the module's RF can still receive network information.

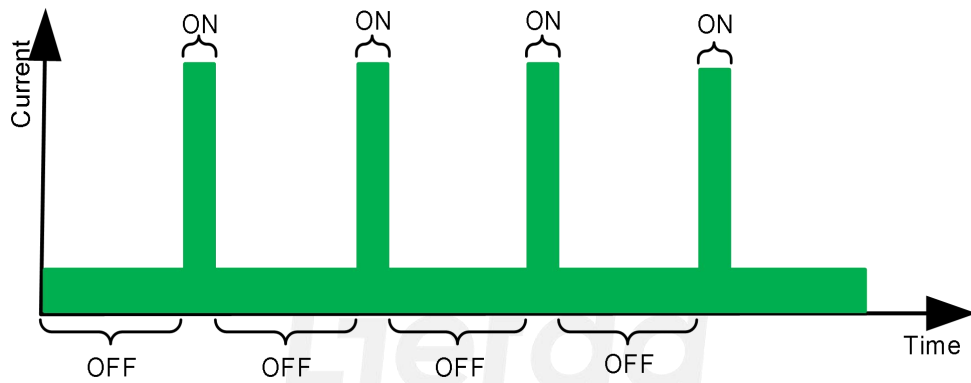


Figure 3.1 Power Consumption States of the Module in Sleep Mode

In sleep mode, the module peripherals are powered down to reduce power consumption, but the RF remains in an intermittent reception state and can still receive network information. When the module has a URC to report, it will wake up the host through the MAIN_RI pin.

3.2.1 Main Serial Port Application Scenarios

If the module and the host communicate via UART, the following two conditions must be met simultaneously for the module to enter sleep mode:

- ◆ Execute the AT+QSCLK=1/2/3/4 command to enable the sleep function;
- ◆ Ensure MAIN_DTR remains at a high level or is left floating.

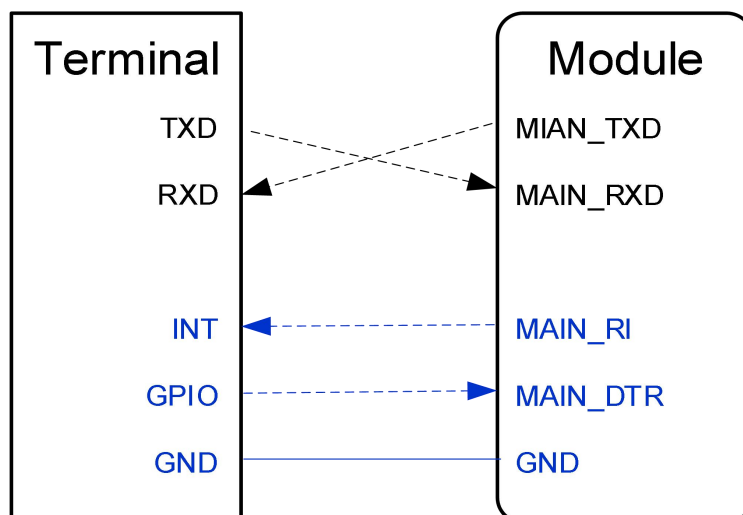


Figure 3.2 Application Circuit of Sleep Mode UART

- ◆ When the module needs to report a URC, the module notifies the host through MAIN_RI.
- ◆ The host pulling MAIN_DTR low can wake up the module.
- ◆ If the host continuously pulls MAIN_DTR low, the module can be prevented from entering sleep mode.
- ◆ If the sleep function is enabled using AT+QSCCLK=2/4, the module can be awakened by sending any command through the main UART interface.

3.3 Airplane mode

When the module enters flight mode, the RF function cannot be used, and all AT commands related to RF are inaccessible. The module can be put into flight mode by the following methods:

Software method, can be set by sending the AT+CFUN=<fun> command. The <fun> parameter can be set to 0 or 1.

- ◆ AT+CFUN=0: Minimum functionality mode (turns off RF functions and USIM card).
- ◆ AT+CFUN=1: Full functionality mode (default).

3.4 Power Supply Design

3.4.1 Power Supply Introduction

The NT21-KCN A series module provides two VBAT pins for connecting external power sources. Additionally, VDD_EXT is the module's external power output and can be used as a reference voltage. There are two hardware versions of VDD_EXT in sleep mode: one with power-off and one without power-off, with power-off being the default. The interface description is as follows:

Table 3-2 Power Pin Definitions

Pin number	Pin name	Description	Minimum value	Typical value	Maximum value	Unit
18,19	VBAT	Modular power supply	3.3	3.8	4.5	V
47	VDD_EXT	Output power	-	1.8	-	V

The module's VBAT power supply range is 3.3 to 4.5V. It is necessary to ensure that the voltage drop caused by the module's operation does not cause the minimum input voltage to fall below 3.3V. The diagram below illustrates the VBAT voltage drop situation during burst transmission of the module:

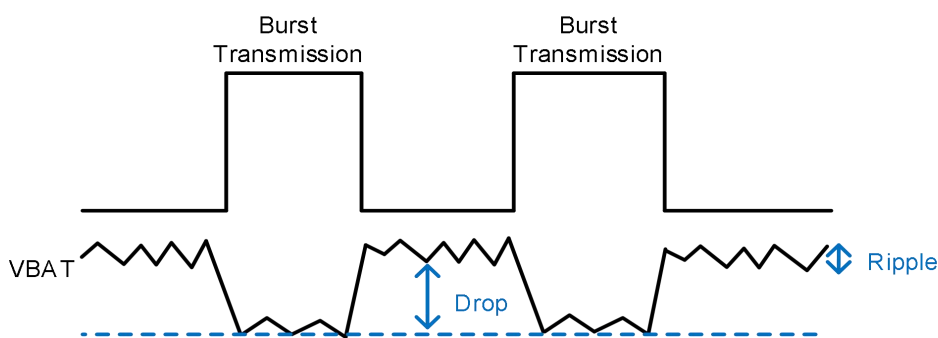


Figure 3.3 Power Requirements for Burst Transmission

Remarks

In sleep mode, VDD_EXT will be powered off.

3.4.2 Power Supply Design

When the module operates at the maximum transmission power of the 4G network, the transient operating current in the live network can reach 1.2A, which may cause a drop in the power supply voltage. Under any circumstances, it is necessary to ensure that the module power supply voltage remains above 3.3V; otherwise, the module may experience unexpected issues such as rebooting. To reduce voltage drops, it is recommended to use a 100uF filter capacitor with low ESR ($ESR < 0.7\Omega$). It is also recommended to reserve three chip-type multilayer ceramic capacitors (MLCC) with good ESR performance for VBAT, and the capacitors should be placed close to the VBAT pin. Additionally, it is advised to add a high-power TVS diode with low clamping voltage and high reverse pulse current (I_{pp}) at the VBAT terminal to enhance the module's surge voltage tolerance.

The power supply reference circuit is as follows:

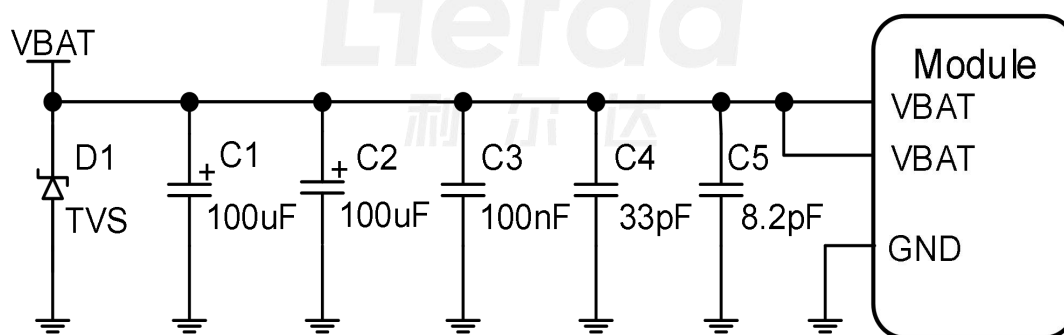


Figure 3.4 Module Power Supply Circuit Reference

In PCB design, the longer the VBAT trace, the wider the trace should be. It is recommended that the trace width be no less than 1mm. The GND plane in the power section should be as complete as possible with multiple ground vias. Additionally, decoupling capacitors should be placed as close as possible to the module's VBAT pin. Among them:

- ◆ D1 is a high-power TVS diode with a low clamping voltage. Due to the small size of the module and insufficient surge protection capability, the TVS diode cannot be omitted;

◆ C1 and C2 are 100uF low ESR tantalum capacitors, enhancing the power supply's current sustainment capability and stabilizing the voltage;

◆ C3~C5 are ceramic package filter capacitors of 100nF, 33pF, and 8.2pF respectively, used to eliminate high-frequency interference.

3.5 Power on

3.5.1 Startup Introduction

The module can be powered on through the PWRKEY pin, with the pin description as follows:

Table 3-3 PWRKEY Pin Description

Pin number	Pin name	I/O	Description	Remarks
1	PWRKEY	DI	Module powers on, active low.	Default is high level.

The control logic of PWRKEY is as follows:

◆ The module can be powered on by pulling the PWRKEY low for at least 500ms while the module is off;

3.5.2 Power-On Circuit Reference Design

It is recommended to use an open-collector driver circuit or directly control power-on through a button. To prevent electrostatic shock caused by contact, a TVS should be placed near the button for ESD protection. The reference circuit is as follows:

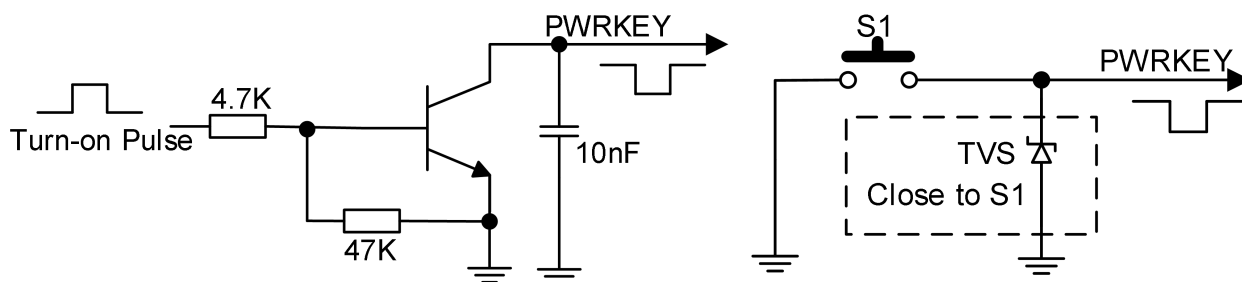


Figure 3.5 Power On/Off Circuit

3.5.3 Power-on sequence

The module power-on startup sequence is shown in the following figure:

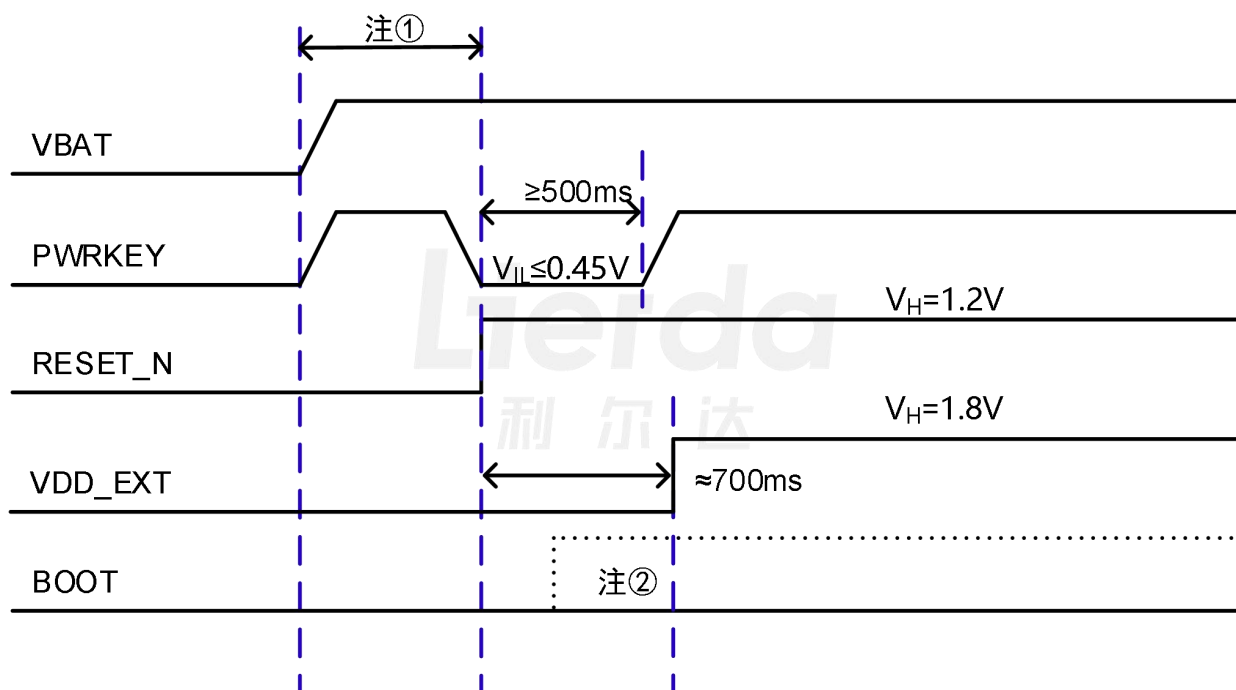


Figure 3.6 Power-On Startup Timing Diagram

Precautions

◆ Note ① The initial state of VBAT power-on must be less than 0.1V, and the time for VBAT to rise from 0V to 2.5V should be less than 10ms; before pulling down the PWRKEY pin, ensure that the VBAT voltage is stable. It is recommended that the time interval between VBAT power-on and pulling down the PWRKEY pin be no less than 30ms.

◆ Note ②: The BOOT pin should be pulled up to VO_LDOIO before the module

powers on; the module will enter emergency download mode when powered on.

◆ If automatic power-on is required without the need for a shutdown function, the PWRKEY can be directly pulled down to ground. It is recommended to use a 4.7kΩ pull-down resistor, or control the PWRKEY to a low level using a GPIO before powering on the module and keep it low continuously.

◆ If there is a low power consumption requirement in the application scenario where PWRKEY is directly pulled down to ground, the AT command "AT+ECPCFG="pwrKeyMode",2" can be used to change the PWRKEY pull-up mode to no pull-up.

◆ The main module serial port sends the "RDY" character, which indicates that the startup is complete.

3.6 Shut down

The module can be properly shut down through the following methods:

- ◆ Control the module shutdown through the PWRKEY pin.
- ◆ Send the AT+QPOWD command to power off.

3.6.1 PWRKEY shutdown

When powered on, pull the PWRKEY low for 650ms continuously and then release it.

The module will execute the shutdown process with the following timing sequence:

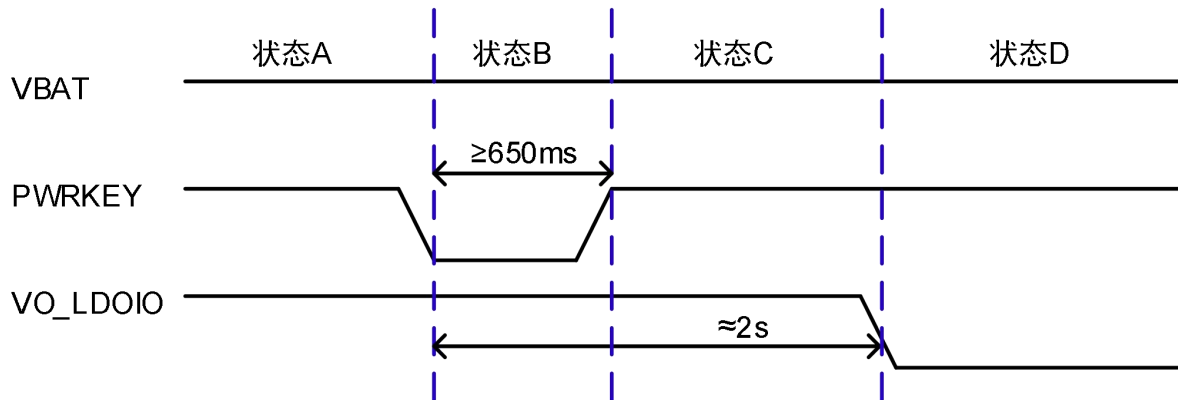


Figure 3.7 PWRKEY Shutdown Timing Diagram

Precautions

- ◆ State A: Module operating status;
- ◆ State B: Lower the PWRKEY state for at least 650ms;
- ◆ State C: The module performs a system shutdown internally, with the time from pulling PWRKEY low to shutdown completion being approximately 2 seconds;
- ◆ State D: The module has entered the shutdown state;

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3.6.2 AT command shutdown

Executing AT+QPOWD can shut down the module, and this operation has the same timing and effect as pulling PWRKEY low to power off.

Precautions

- ◆ When the module is operating normally, do not immediately cut off the module's power, as this may damage the internal Flash data of the module. It is recommended to first shut down the module using the PWRKEY or AT command before disconnecting the power.
- ◆ When using AT commands to shut down, please ensure that the PWRKEY remains at a high level after the shutdown command is executed; otherwise, the device may

restart after shutdown, and disconnecting the power at this time risks damaging the Flash.

◆ If using the method of keeping PWRKEY commonly grounded to power on, it is recommended to use AT+QPOWD to execute the shutdown process before disconnecting the power.

3.7 Reset

3.7.1 Reset Introduction

Pulling RESET_N low for at least 300 ms and then releasing it can reset the module.

The description of the RESET_N pin is as follows:

Table 3-4 Reset Pin Description

Pin number	Pin name	I/O	Description	DC Characteristics	Remarks
16	RESET_N	DI	Module Reset	VILmax=0.3V	There is a weak pull-up inside, active low, leave floating if not used.

3.7.2 Reset Circuit Reference Design

The RESET_N signal is relatively sensitive to interference. It is recommended that the module interface wiring be as short as possible and should be grounded. The typical reference hardware circuit application is as follows, with two circuit diagrams available for selection based on actual requirements:

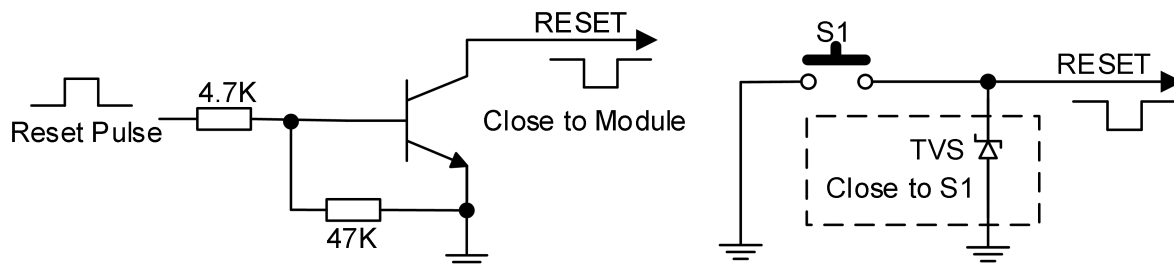


Figure 3.8 Reset Circuit Reference Design

3.7.3 Reset timing

The reset timing sequence of the module is as follows:

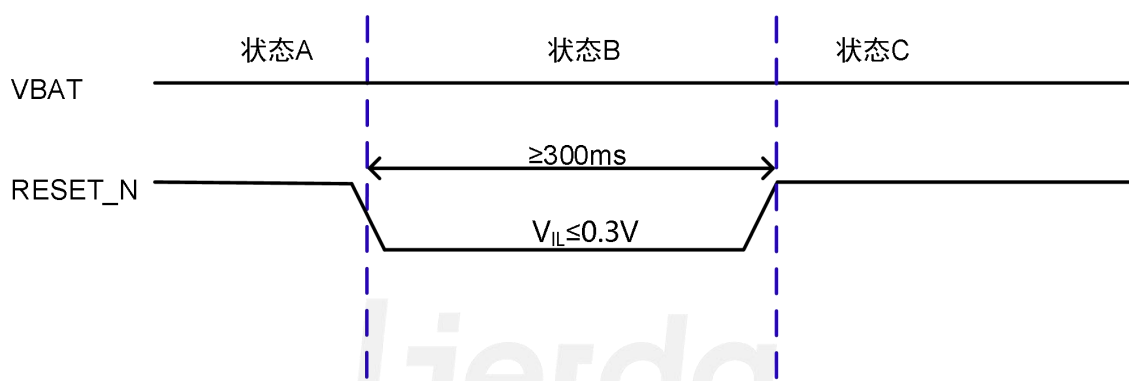


Figure 3.9 Reset Timing

Precautions

- ◆ State A: Module normal working condition;
- ◆ State B: Pull the RESET_N line low for at least 300ms;
- ◆ State C: The module begins to restart;

4 Application Interface

The NT21-KCN A series modules offer a wide range of peripheral functions, mainly including the following application interfaces:

- ◆ USB 2.0 Interface
- ◆ UART Interface
- ◆ USIM Interface
- ◆ I2C Interface
- ◆ Network Status Indicator Interface

4.1 USB interface

The NT21-KCN A series module supports a USB 2.0 interface, complies with USB 2.0 specifications, and supports full-speed (12 Mbps) and high-speed (480 Mbps) modes. The module only supports USB slave mode. This interface can be used for AT command transmission, data transfer, software debugging, and firmware upgrades. The interface definition is as follows:

Table 4-1 USB Interface Pin Description

Pin number	Pin name	I/O	Description	DC Characteristics	Remarks
6	VBUS	DI	USB plug-in wake-up module	Vmax=5.25V Vmin=3.0V Vnom=5.0V	In sleep mode, USB insertion will wake up the module.
25	BOOT	DI	Emergency download mode control. Before the module powers on, pull up to VO_LDOIO; after the module starts, it will enter emergency download mode.	VO_LDOIO	The module has a default weak pull-down internally and can be used as a GPIO after power-on; if not used, it should be left floating.

7	USB_DP	DIO	USB differential data (+)	-	If not used, leave floating
8	USB_DM	DIO	USB differential data (-)	-	If not used, leave floating

4.1.1 USB Circuit Reference Design

The module USB interface can be connected to a USB connector or the MCU's USB interface. The reference schematic for connecting to the USB connector is as follows:

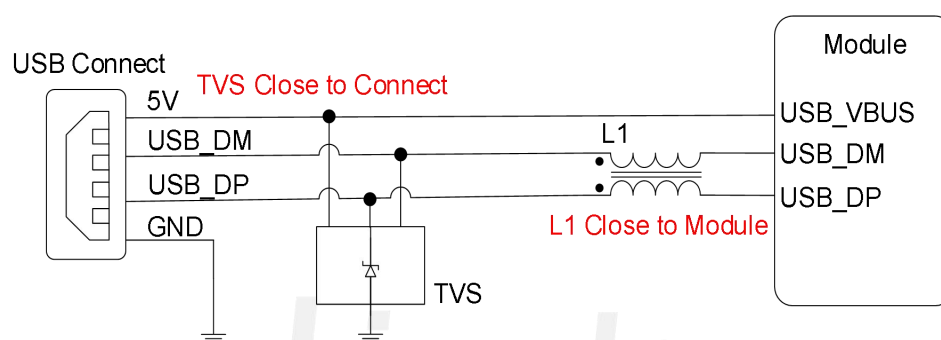


Figure 4.1 USB Interface Reference Design

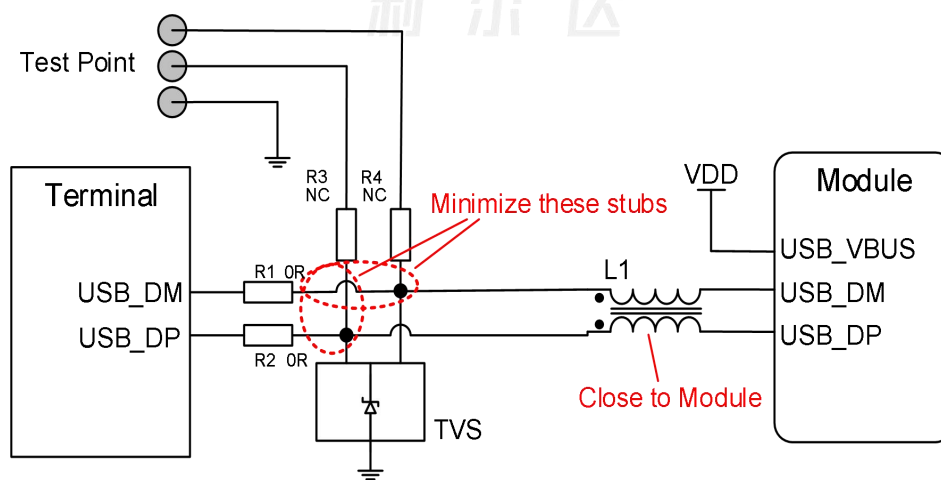


Figure 4.2 USB Connection MCU Reference Design

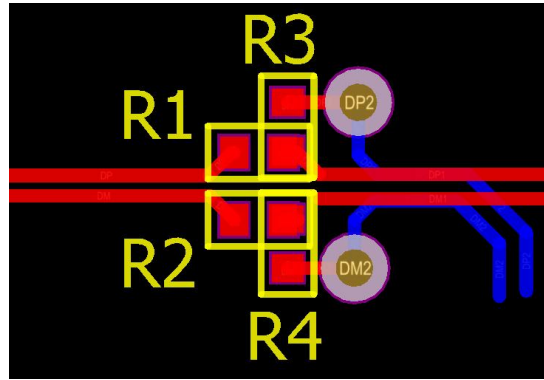


Figure 4.3 USB Jumper Routing Diagram

Precautions

◆ In order to ensure USB performance, the USB_DP and USB_DM lines should maintain a $90\ \Omega$ differential impedance, with grounding treatment around them. It is recommended to connect a common mode choke L1 in series between the MCU and the module to prevent EMI interference from the USB signal. L1 should be placed close to the module.

◆ USB wiring should be routed away from crystal oscillators, oscillators, magnetic devices, RF signals, and strong signal areas. It is recommended to use inner layer differential routing with three-dimensional ground shielding on all sides (top, bottom, left, and right).

◆ If the module's USB interface and MCU are connected through a connector, a TVS diode should be added at the interface for protection. When selecting ESD protection devices, special attention should be paid to ensure their parasitic capacitance does not exceed 2pF, and they should be placed as close to the USB interface as possible.

◆ Reserve resistors R1~R4. Normally, R1~R2 are mounted and R3~R4 are not, allowing normal connection to the MCU; when debugging is needed, R3~R4 can be mounted and R1~R2 removed to switch to the test points. To meet the USB data line signal integrity requirements, the branch lines connecting to the test points should be kept as short as possible.

4.1.2 USB Data Transfer

When the module is powered on or reset, if BOOT is detected to be at a low level, the USB port will enumerate two serial ports for sending AT commands and log output.

4.1.3 USB firmware download

When the module is powered on or reset/restarted, if BOOT is detected to be at a high level, the USB port will enumerate a download-specific serial port. At this time, the download tool can perform firmware upgrades through the USB interface. If no download tool is connected within 15 seconds, the module will re-enumerate two serial ports according to the default USB configuration.

The BOOT interface reference design is as follows:

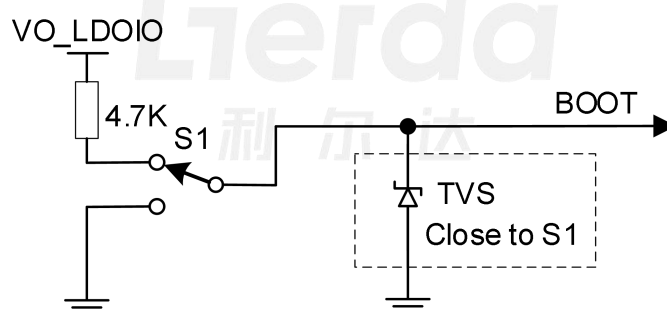


Figure 4.4 BOOT Reference Design

4.2 UART Communication

The module provides three general asynchronous transceivers: the main serial port, the debug serial port, and the auxiliary serial port. The baud rate of the main serial port is configurable, while the debug serial port is only used for debugging and testing purposes. The serial port pin definitions are as follows:

Table 4-2 Serial Port Pin Definitions

Interface	Pin number	Pin name	Description	Remarks
Main serial port	14	MAIN_TXD	Main serial port sends data	If not used, leave floating
	15	MAIN_RXD	Main serial port receiving data	Non-sleep state: VO_LDOIO power domain; Sleep state: switch to internal VDD18AON power domain
	5	MAIN_RI	Main serial port outputs ring indication	If not used, leave floating
	50	MAIN_DTR	Main serial port data terminal ready	In sleep mode, it can serve as a wake-up pin.
Debug serial port	2	DBG_RXD	Debugging serial port data reception	If not used, leave floating
	3	DBG_TXD	Debug serial port data transmission	If not used, leave floating
Auxiliary serial port	31	AUX_TXD*	Assist with serial port data transmission	Not supported yet
	32	AUX_RXD*	Assist with serial port data reception	Not supported yet

Precautions

- ◆ The serial port interface belongs to the VO_LDOIO power domain, with a default voltage level of 1.8V;
- ◆ The VO_LDOIO power domain will power down in sleep mode, and MAIN_RXD switches to the internal 1.2V power supply;
- ◆ The RXD pin of the serial port has an internal pull-up resistor, so no external pull-up resistor is needed.
- ◆ Pay attention to the consistency of voltage levels when using the serial port,

otherwise leakage current may easily occur;

4.2.1 Serial port application

Main characteristics of the primary serial port:

- ◆ Used for AT command communication and data transmission, with a default baud rate of 115200bps;
- ◆ Configurable to: 4800/9600/19200/38400/57600/115200/230400/460800bps;
- ◆ Used for firmware upgrade, the default baud rate during upgrade is 921600bps;
- ◆ Supports low-power wake-up module functionality.

Debugging serial port characteristics:

- ◆ Through the dedicated tools provided by the platform, log information can be viewed for software debugging, with a default rate of 3000000bps.

The schematic diagram of the serial port connection method is as follows:

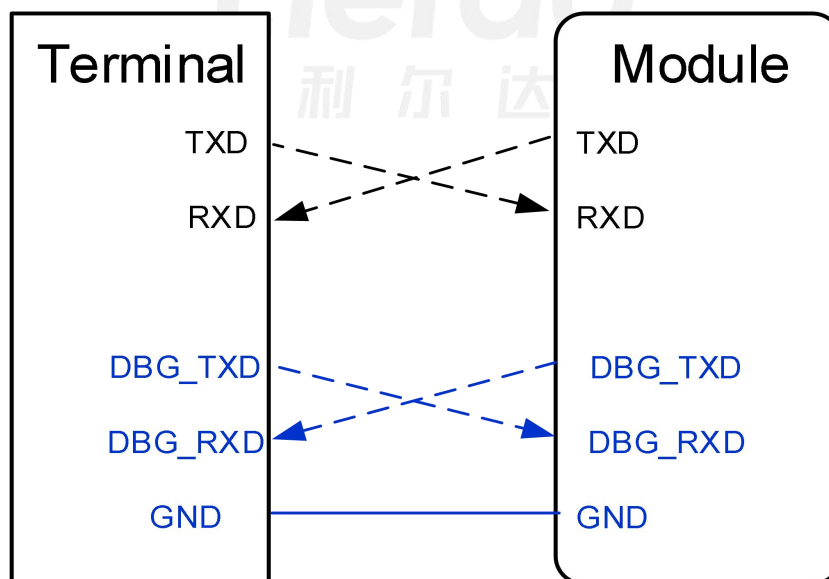


Figure 4.5 Serial Port Connection Diagram

4.2.2 Serial Port Circuit Reference Design

The main factors to consider for a suitable serial port level conversion circuit include whether it meets the operating speed of the serial port, whether it is suitable for low power

consumption scenarios, and whether its power consumption meets the requirements. It is recommended to refer to the following level matching circuit:

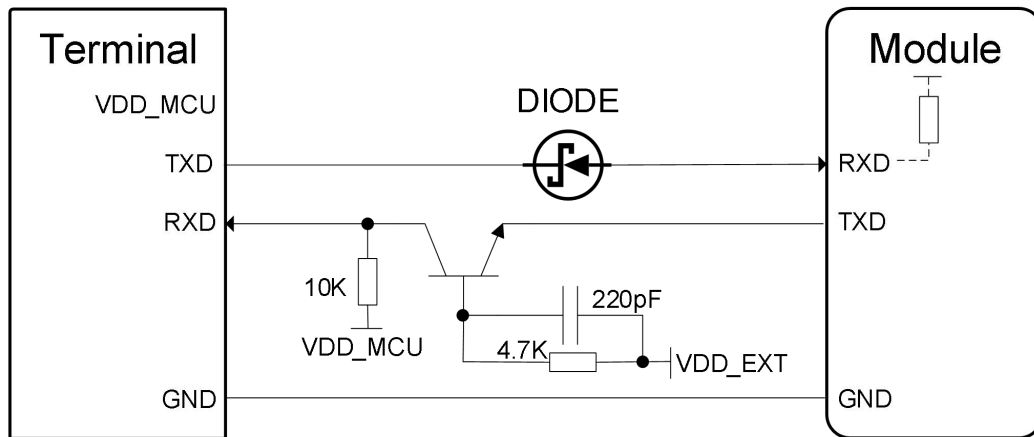


Figure 4.6 Transistor Level Conversion Reference Circuit

Recommended transistors for reference:

Brand: CJ Specification Model: S8050 J3Y Package: SOT-23

Recommended diodes for reference (note that the forward voltage of the diode should be $\leq 0.3V$):

Brand: LRC Specification Model: LRB520S-30T1G Package: SOD-523

4.3 USIM card interface

The module includes one external USIM card interface, compliant with ETSI and IMT-2000 standards, supporting 1.8 V and 3.0 V USIM cards. The pin description is as follows:

Table 4-3 External USIM Card Interface Pin Definitions

Pin number	Pin name	Description	DC Characteristics
10	SIM_RST	SIM card reset line	USIM_VDD
11	SIM_CLK	SIM card clock line	USIM_VDD
12	SIM_DATA	SIM card data cable	USIM_VDD

13	SIM_VDD	SIM card power supply	Vnorm=1.8/3.0V
48	SIM_DET	SIM Card Status Signal Detection	SIM card slot hot-plug detection interface

The module supports hot swapping of USIM cards through SIM_DET, which is active high by default and can be configured for high or low level detection. If the USIM card detection function is not needed, SIM_DET can be left floating.

4.3.1 USIM Card Circuit Reference Design

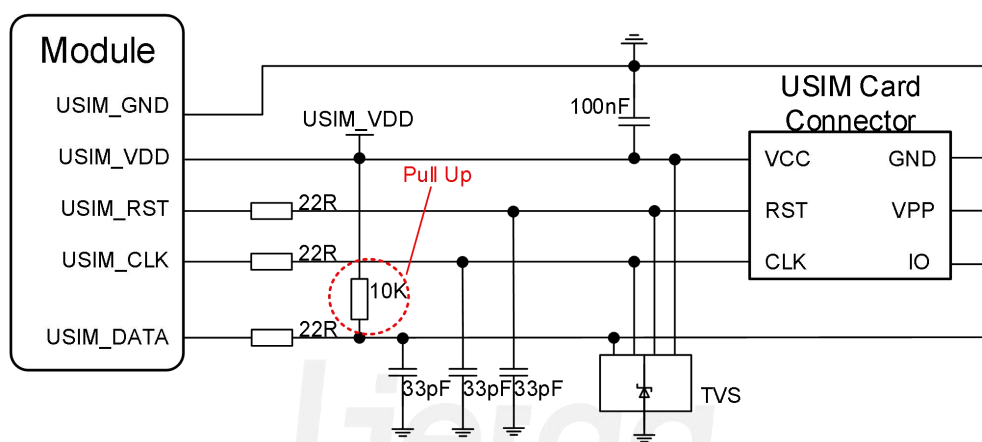


Figure 4.7 6PIN SIM Card Reference Circuit

4.3.2 Precautions for USIM Card Circuit Design

To ensure the reliability and availability of the SIM card in the application, please follow the following USIM circuit design guidelines:

- ◆ When laying out, try to place the SIM card as close to the module as possible, and the signal line wiring length should preferably not exceed 200mm;
- ◆ The SIM card signal line should be kept away from the RF routing and VBAT power line;
- ◆ The GND wiring of the SIM card should be short and thick, ensuring the wiring width is not less than 0.5mm;
- ◆ The decoupling capacitor for USIM_VDD should not exceed 1uF, and the capacitor

should be placed close to the VCC of the external SIM card;

◆ To avoid signal crosstalk between USIM_DATA and USIM_CLK, their routing should not be too close. A ground shield must be added between the two traces. Additionally, to prevent the effects of long routing, USIM_DATA generally requires a pull-up resistor to USIM_VDD to enhance driving capability. If the routing is too long, it is recommended to externally reserve a 10K resistor placed near the card slot. Furthermore, the USIM_RST signal also needs to be shielded by ground protection.

◆ To ensure good ESD protection performance, it is recommended to add TVS diodes to the pins of the external USIM card. The parasitic capacitance of the selected TVS diode should not exceed 15pF; a 22-ohm resistor should be connected in series on the signal lines between the module and the SIM card to suppress stray EMI and enhance ESD protection. Additionally, a parallel 33pF capacitor is used to filter out RF interference. The related resistors, capacitors, and TVS diodes should be placed close to the USIM card socket.

4.4 I2C interface

The module provides one I2C interface, with the interface description as follows:

Table 4-4 I2C Interface Pin Definitions

Pin number	Pin Name	Description	DC Characteristics	Remarks
33	I2C0_SDA	I2C0 Data	VO_LDOIO	External pull-up is required.
49	I2C0_SCL	I2C0 clock	VO_LDOIO	External pull-up is required.
38	I2C1_SDA	I2C1 Data	VO_LDOIO	External pull-up required
37	I2C1_SCL	I2C1 clock	VO_LDOIO	External pull-up is required.

4.5 ADC interface

The module provides one 12-bit analog-to-digital converter input interface (ADC interface). When wiring the ADC interface, to improve the accuracy of voltage measurement, it is recommended to implement ground shielding. The pin definitions are as follows:

Table 4-8 ADC Pin Definitions

Pin number	Pin Name	Description	DC characteristics	Remarks
34	ADC0	General ADC Interface	Internal direct connection Input voltage range: 0~1.05V	If not used, leave floating
35	ADC1	General ADC Interface	Internal voltage division Input voltage range: 0~3.3V Default 1.05V The internal voltage division can be enabled through software.	If not used, leave floating

Precautions

◆ If the sampling voltage is greater than 1.05V, it is recommended to use a resistor voltage divider circuit for the ADC pin input. The resistance value of the voltage divider resistors should not exceed 100 k Ω , otherwise it will reduce the measurement accuracy of the ADC.

◆ Reserve a 1nF capacitor across the grounding voltage divider resistor during design, default not to mount.

4.6 Network status indicator signal

The module supports network status, with the interface defined as follows:

Table 4-9 Status Indicator Pin Definitions

Pin number	Pin Name	Description	DC characteristics	Remarks
27	NET_STATUS	Network Status Indicator	VDD18AON	If not used, leave floating

The NET_STATUS pin is used to indicate the network status of the module. The logic levels for different states are as follows:

Table 4-10 Status Indicator Pin Definitions

Pin Name	Level logic	Network status
NET_STATUS	Slow flash (200ms high / 1800ms low)	Network search status
	Slow flash (1800ms high / 200ms low)	Standby mode
	Flash (125ms high / 125ms low)	Data Transmission Mode

4.6.1 Indicator Light Circuit Reference Design

NET_STATUS can be used to drive the LED indicator light. The reference design is as follows:

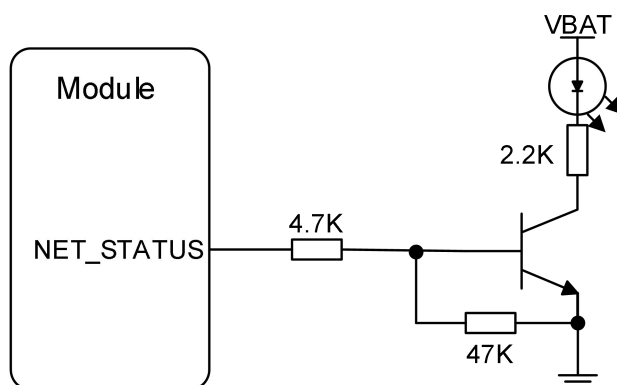


Figure 4.9 Indicator Light Circuit Reference Circuit



5Radio Frequency Characteristics

5.1 Introduction to RF Functions

The NT21-KCN A series module provides one RF interface with a characteristic impedance of 50Ω. The interface description is as follows:

Table 5-1 RF Interface Pin Definitions

Pin number	Pin Name	I/O	Description	DC Characteristics	Remarks
23	LTE_ANT	AIO	Main antenna interface	-	Used for over-the-air data transmission

5.2 Cellular network

5.2.1 Operating frequency band

The NT21-KCN A series module supports the following operating frequency bands:

Table 5-2 Operating Frequency Bands

Frequency band Band	Uplink frequency band Uplink(UL)band	Downlink frequency band Downlink(DL)band	Network standard Duplex Mode
3	1710MHz-1785MHz	1805MHz-1880MHz	FDD
8	880MHz-915MHz	925MHz-960MHz	FDD
34	2010MHz-2025MHz	2010MHz-2025MHz	TDD
38	2570MHz-2620MHz	2570MHz-2620MHz	TDD
39	1880MHz-1920MHz	1880MHz-1920MHz	TDD
40	2300MHz-2400MHz	2300MHz-2400MHz	TDD
41	2496MHz-2690MHz	2496MHz-2690MHz	TDD

5.2.2 Transmission Power

The transmission power of the NT21-KCN A series module is as follows:

Table 5-3 Conducted Power

Band	Network standard	Maximum value	Minimum value	Remarks
3	FDD	23dBm±2.7dB	<-40dBm	Compliant with Cat.1 protocols in 3GPP Rel-13 and Rel-14.
8	FDD	23dBm±2.7dB	<-40dBm	
34	TDD	23dBm±2.7dB	<-40dBm	
38	TDD	23dBm±2.7dB	<-40dBm	
39	TDD	23dBm±2.7dB	<-40dBm	
40	TDD	23dBm±2.7dB	<-40dBm	
41	TDD	23dBm±2.7dB	<-40dBm	

5.2.3 Receiving sensitivity

The reception sensitivity of the NT21-KCN A series module is as follows:

Table 5-4 Conduction Reception Sensitivity under Single Transmission (Throughput ≥ 95%)

Band	Network standard	Reception Sensitivity (10M)	3GPP Standard (10M)
3	FDD	-98.5	-93.3dBm
8	FDD	-99.5	-93.3dBm
34	TDD	-99.8	-96.3dBm
38	TDD	-99.3	-96.3dBm
39	TDD	-99.3	-96.3dBm

40	TDD	-99.5	-96.3dBm
41	TDD	-100.1	-94.3dBm

5.3 Antenna Reference Circuit Design

When users are using the NT21-KCN A series module, a π -type matching circuit needs to be added between the module's RF antenna interface and the user's baseboard antenna interface. The typical antenna matching circuit and initial parameters are shown in the figure below:

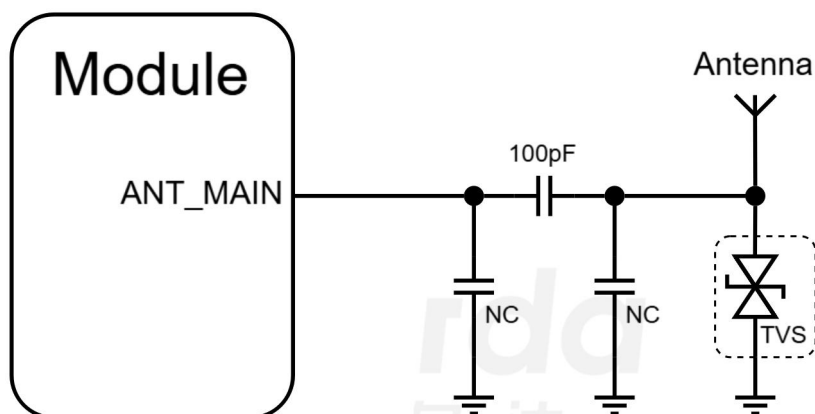


Figure 5.1 RF Antenna Reference Circuit

Precautions

- ◆ The series position uses 100pF capacitors; capacitor positions are left unpopulated by default. The recommended component packages are 0201 or 0402.
- ◆ If an external antenna is used, or if the user can touch the antenna, it is recommended to reserve a TVS diode to enhance electrostatic protection. Since the parasitic capacitance of the TVS may affect antenna performance, a model with parasitic capacitance $\leq 0.5\text{pF}$ should be selected. After adding the TVS diode, it is recommended to retune the antenna.

5.4 Guidelines for RF Signal Cable Routing

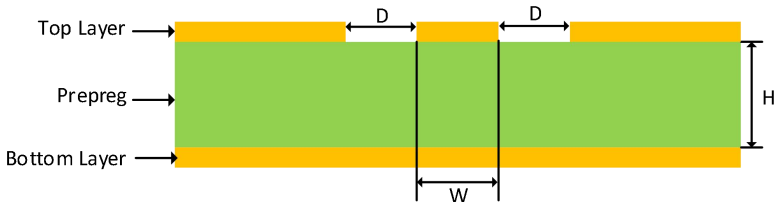
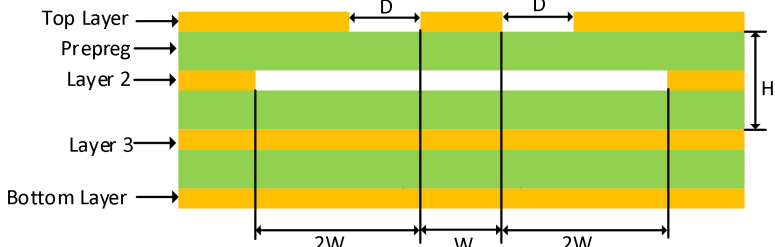
The routing between the module antenna interface and the user antenna must meet the 50-ohm characteristic impedance requirement. At the same time, the RF routing should be as short as possible to ensure minimal insertion loss of the RF line.

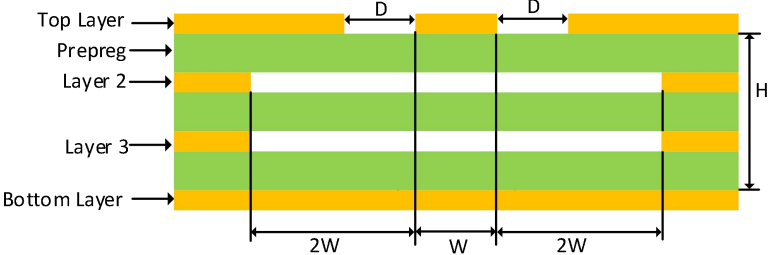
Under normal circumstances, the impedance of an RF signal line is determined by the dielectric constant ϵ_r of the material, the trace width W , the distance to ground D , and the thickness H of the reference ground plane.

In the field of Internet of Things applications, the design of PCB characteristic impedance is usually implemented using a coplanar waveguide method, which helps achieve better shielding of RF signal lines while enabling higher integration for compact area design.

Common PCB coplanar waveguide designs include the following types:

Table 5-5 Design Instructions for Common PCB Coplanar Waveguide Structures

Serial Number	Pin Name	Remarks
1		2-layer board The reference ground is the L2 layer.
2		4-layer board The reference point is the L3 layer. L2 layer hollowing

3		4-layer board The reference point is at layer L3. L2 and L3 layer hollowing
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During the layout process, a 50-ohm impedance can be simulated and calculated using the Polar Si9000 tool. Based on the actual situation, select the appropriate line width, line spacing, and stacking to achieve the best design results.

Take the calculation example of a 2-layer board, 1.6mm thickness PCB coplanar waveguide:

- ◆ Select the appropriate PCB substrate material. Once the substrate is chosen, the dielectric constant ϵ_r is basically determined (in this case, 4.6);
- ◆ Choose an appropriate trace width. A good method is to select a trace width that matches the size of the pads of the components being matched (note that it should not be too thin; in this example, 0.65mm is chosen);
- ◆ Based on the selected board thickness, line width, dielectric constant, etc., the spacing to ground can be roughly calculated. If the line spacing is not suitable, the line width can be adjusted to modify the line spacing (calculated to be 0.14mm, which is basically appropriate).



Figure 5.2 Reference for the calculation method of 50-ohm impedance

During the actual layout process, the following suggestions are provided for reference:

- ◆ In the π -type circuit, the three matched reserved components are placed close to the antenna in a compact arrangement, and the bypass components are recommended to be placed on both sides of the RF line.
- ◆ Place irregular vias on both sides of the RF trace's GND plane; there must be a complete GND plane beneath the entire RF trace area.
- ◆ There should be no other wiring below the RF line to avoid affecting RF performance or other circuits.

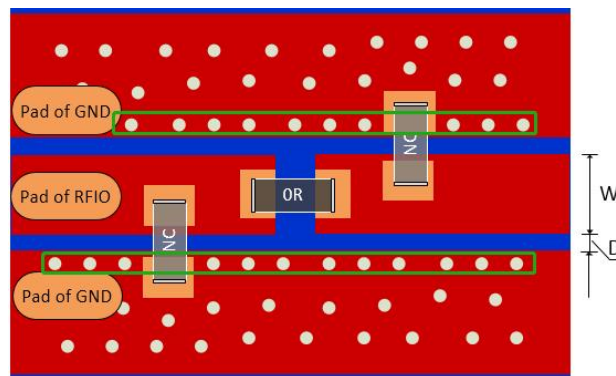


Figure 5-3 RF Trace Layout Design Diagram

The rationality of the RF routing can be directly reflected in the module's conduction test. To ensure the product's overall performance is maximized, antenna design coordination is also required. To better meet the needs of antenna design, the following requirements are hoped to be achieved in PCB design. Below are guidelines for the whole machine PCB with different numbers of layers:

- ◆ When the product PCB is designed as a 2-layer board, it is best for the TOP and BOTTOM LAYER directly under the module to both be GND layers. The routing that needs to be led out from the module should avoid passing directly underneath the module and should be routed from the outside of the module.
- ◆ When the product PCB is designed with 4 layers, it is recommended that the module's signal traces be routed on the third or fourth layer, reserving the first and second

layers as complete GND reference layers for the module.

5.5 Antenna Design Requirements

The RF performance of the module is also affected by the antenna. The selection of the antenna needs to meet the following requirements:

- ◆ Select an antenna that matches the module's operating frequency band;
- ◆ The antenna's characteristic impedance is required to be 50 ohms to reduce the loss at the connection between the RF line and the antenna;
- ◆ The insertion loss within the operating frequency band should be as low as possible, such as a standing wave ratio (VSWR) ≤ 2 and return loss ≥ 10 dB.

Commonly used antennas suitable for CAT1 scenarios are shown in the figure below:



Figure 5.4 Common Antenna Types for CAT1

5.6 Recommended RF Connectors

In some applications that require customized antennas, such as environments where the device is shielded by an iron case, an IPX or SMA connector can be reserved on the motherboard, allowing the antenna to be connected outside the device via an RF cable to achieve better performance.

Table 5.6 Common RF Connectors

Serial	Product Name	Image	Remarks
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Number			
1	IPX socket		Connect the RF cable
2	SMA connector		Can connect to RF cable or antenna.

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6Electrical Performance and Reliability

6.1 Absolute maximum rating

The absolute maximum rated values of the module interface are described as follows:

Table 6-1 Absolute Maximum Ratings

Parameters	Description	Minimum value	Maximum value	Unit	Remarks
V _{BAT}	Module power supply voltage	-0.3	5.0	V	
V _{IAIO}	ADC input voltage	-0.3	3.6	V	
V _{IGPIO}	GPIO input voltage	-0.3	3.6	V	
Others	Other pin input voltage	-0.3	3.6	V	

Precautions

Permanent damage to the module may occur if the conditions exceed the absolute maximum ratings.

6.2 Power rating

The electrical characteristics of the module power interface are described as follows:

Table 6-2 Electrical Characteristics of Power Interface

Parameters	Description	Minimum value	Typical value	Maximum value	Unit
V _{BAT}	Module normal power supply voltage (1)	3.3	3.8	4.5	V
V _{BAT}	Module expansion power supply voltage (2)	3.1	3.8	4.5	V
USIM_VDD	USIM card power supply output	-	1.8/3.0	-	V
VDD_EXT	Reference power output (3)	-	1.8	-	V

In the sleep mode VDD_EXT power-off version, the normal driving capability is 120mA, with no output in sleep mode.

In sleep mode, the VDD_EXT non-power-off version has a driving capability of 3 to 5mA.

Precautions

◆ Note (1): When the module operates within this voltage range, the relevant performance of the module meets the requirements of the 3GPP standard.

◆ Note (2): When the module operates within this voltage range, it can still maintain normal operation without any irreversible faults; only certain parameters, such as output power, may exceed the range specified by the 3GPP standard. When the voltage returns to the normal operating range, all module parameters will still comply with the 3GPP standard.

◆ Note (3): VDD_EXT will lose power in sleep mode.

6.3 Power consumption

The power consumption of the module is as follows:

Table 6-3 Module Power Consumption

Module description	Test conditions	Typical value	Unit
Shutdown mode	Module shutdown	0.85	uA
Sleep mode	AT+CFUN=0 (USB disconnected)	6	uA
	LTE-FDD @ PF = 32 (USB disconnected)	0.81	mA
	LTE-FDD @ PF = 64 (USB disconnected)	0.58	mA
	LTE-FDD @ PF = 128 (USB disconnected)	0.19	mA
	LTE-FDD @ PF = 256 (USB disconnected)	0.14	mA
	LTE-TDD @ PF = 32 (USB disconnected)	0.83	mA
	LTE-TDD @ PF = 64 (USB disconnected)	0.61	mA
	LTE-TDD @ PF = 128 (USB disconnected)	0.19	mA

	LTE-TDD @ PF = 256 (USB disconnected)	0.15	mA
Idle mode	LTE-FDD @ PF = 64 (USB disconnected)	3.25	mA
	LTE-TDD @ PF = 64 (USB disconnected)	3.25	mA
LTE Data Transmission	LTE-FDD B3	611	mA
	LTE-TDD B8	544	mA
	LTE-TDD B34	215	mA
	LTE-TDD B38	248	mA
	LTE-TDD B39	237	mA
	LTE-TDD B40	243	mA
	LTE-TDD B41	251	mA

6.4 Digital Logic Level Characteristics

The logic level definitions for the module digital IO are as follows:

Table 6-4 Digital IO Logic Level Description

Type	Parameters	Description	Minimum value	Maximum value	Unit
Input	VIL	Input low level	-	0.2*VDD_EXT	V
	VIH	Input high level	0.7*VDD_EXT	-	V
Output	VOL	Output low level	-	0.15*VDD_EXT	V
	VOH	Output high level	0.8*VDD_EXT	-	V

Precautions

The module's GPIO reference level follows VDD_EXT. When VDD_EXT powers down during sleep, the corresponding GPIO will also power down.

The logic level definitions for the module USIM card interface are as follows:

Table 6-5 USIM Logic Level Description

Type	Parameter	Description	Minimum value	Maximum	Unit
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	s			value	
Input	VIL	Input low level	-	$0.2 \times \text{USIM_VDD}$	V
	VIH	Input high level	$0.7 \times \text{USIM_VDD}$	-	V
Output	VOL	Output low level	-	$0.15 \times \text{USIM_VDD}$	V
	VOH	Output high level	$0.8 \times \text{USIM_VDD}$	-	V

Precautions

The USIM_VDD voltage level is determined based on the detected type of USIM card and supports 1.8V/3.0V USIM cards.

6.5 Electrostatic Protection

In daily life and production, static electricity is everywhere. Static electricity generated by the human body and friction between objects can be transmitted to modules through various means and may potentially damage the modules. Therefore, it is essential to pay close attention to static electricity protection and take necessary anti-static measures. For example:

- ◆ Wear anti-static gloves during research and development, production, assembly, and testing processes;
- ◆ When designing products, add anti-static protection devices at circuit interfaces and other points susceptible to electrostatic discharge.

The electrostatic discharge performance of the module is as follows:

Table 6-6 Electrostatic Protection Characteristics

Test interface	Contact discharge	Remarks
VBAT	$\pm 8\text{KV}$	Test Standard: IEC61000-4-2 Temperature: 25°C Humidity: 45%
GND	$\pm 8\text{KV}$	
Antenna interface	$\pm 8\text{KV}$	

6.6 Operating and storage temperature

Table 6-7 Operating Temperature Range

Parameters	Minimum value	Typical value	Maximum value	Unit	Remarks
Normal operating temperature (1)	-35	+25	+75	°C	
Extended Operating Temperature (2)	-40	+25	+85	°C	
Storage ambient temperature (3)	-40	+25	+90	°C	

Precautions

(1) When the module operates within this temperature range, the module's related performance meets the 3GPP standard requirements.

(2) When the module operates within this temperature range, it can still maintain normal operation without any irrecoverable failures; only a few parameters, such as output power, may exceed the 3GPP standard range. When the temperature returns to the normal operating range, all parameters of the module still comply with the 3GPP standards.

(3) This storage temperature range does not include packaging materials; attention should be paid to the maximum tolerance temperature of the tape packaging.

6.7 Precautions

There are some precautions to be noted during the production and use of the module:

◆ When spraying the module, please try to avoid the spraying material flowing into the inside of the module;

◆ When cleaning the module, do not use ultrasonic cleaning on the module, as it may cause damage to the internal crystal of the module;

◆ When producing and using the module, please avoid applying it in any environment

or packaging containing mercury or mercury vapor, as this may lead to product failure or malfunction risks.



7 Mechanical dimensions

7.1 Mechanical dimensions

The module mechanical dimension drawing is as follows:

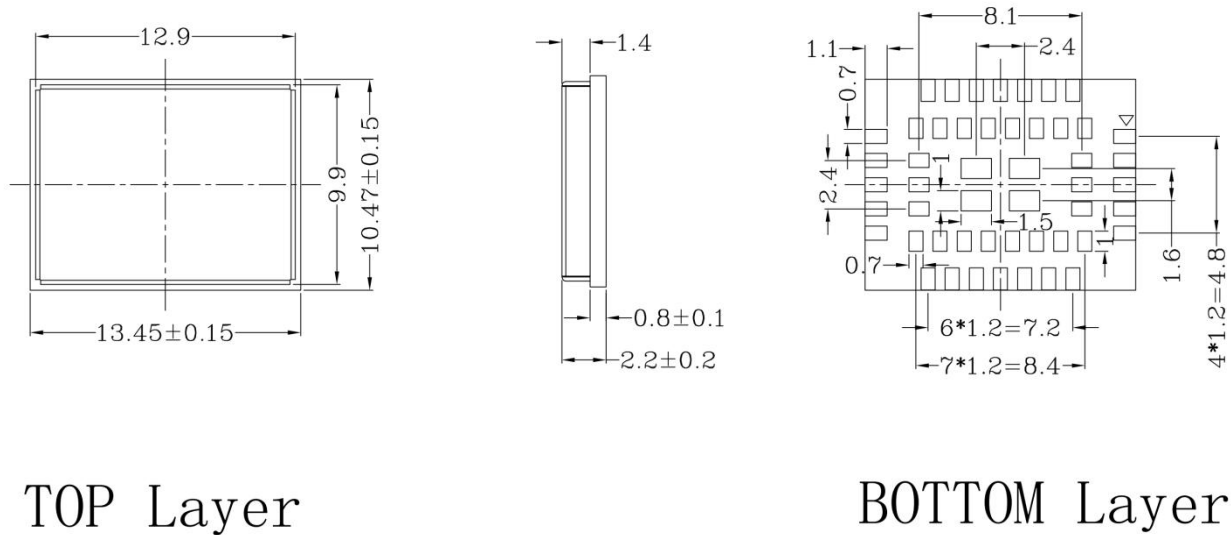


Figure 7.1 Module Mechanical Dimensions (Unit: mm)

7.2 Module Top View/Bottom View

The actual module physical effect diagram is as follows, please refer to the actual module for details:

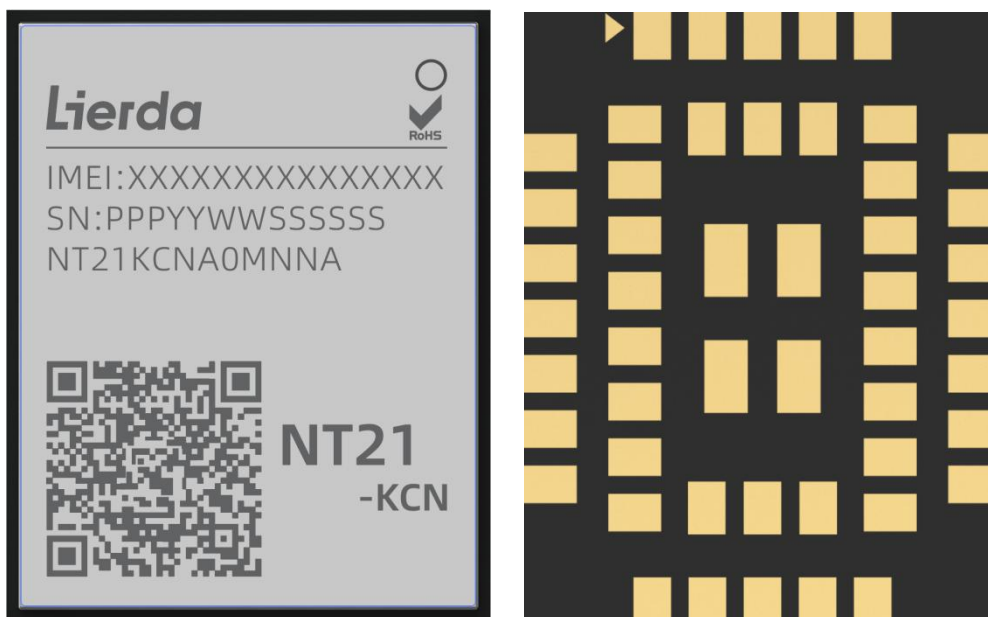


Figure 7.2 Module Top/Bottom View

7.3 Recommended packaging

The recommended module solder pads are shown in the figure below. Users can make fine adjustments according to their own production processes.

◆ The module's surrounding pins feature an internal rounded corner design. When designing the PCB pads, users should consider using rounded corner transitions; for the rectangular pads at the bottom of the module, the PCB design can adopt the module pin dimensions, as shown in the reference design diagram for a single pad below.

◆ To facilitate the opening of stepped stencils, it is recommended not to place other components within 2.0mm outside the module pads. Users can refer to the requirements of their own stencil manufacturers to determine this distance.



This chapter describes the module's surface-mount technology, packaging methods, storage conditions, and other information, which can help users better preserve and use the module.

8.1 Production welding

8.1.1 Furnace processing method

When mounting the module, the following precautions should be observed during the reflow process:

- ◆ If the module's baseboard is double-sided SMT, it is recommended to place the module during the second SMT process;

- ◆ During the first chip mounting, it is best for the customer's substrate to pass through the oven on the conveyor belt; for the second chip mounting, it is also preferable to place it on the conveyor belt for oven processing.

- ◆ If, for special reasons, the PCB cannot be placed on the conveyor belt for reflow, consider using a fixture to pass it through the reflow oven on the track or placing a flat, high-temperature resistant, straight template to support the PCBA during reflow, preventing PCB deformation that could cause poor solder joints in the module.

8.1.2 Reflow Soldering Operation Instructions

The PCBA reflow oven temperature profile is related to the solder paste used and needs to be adjusted according to the actual solder paste. The data is only suitable for lead-free operations. Refer to Figure 8-2 Lead-Free Reflow Soldering Operation Guide.

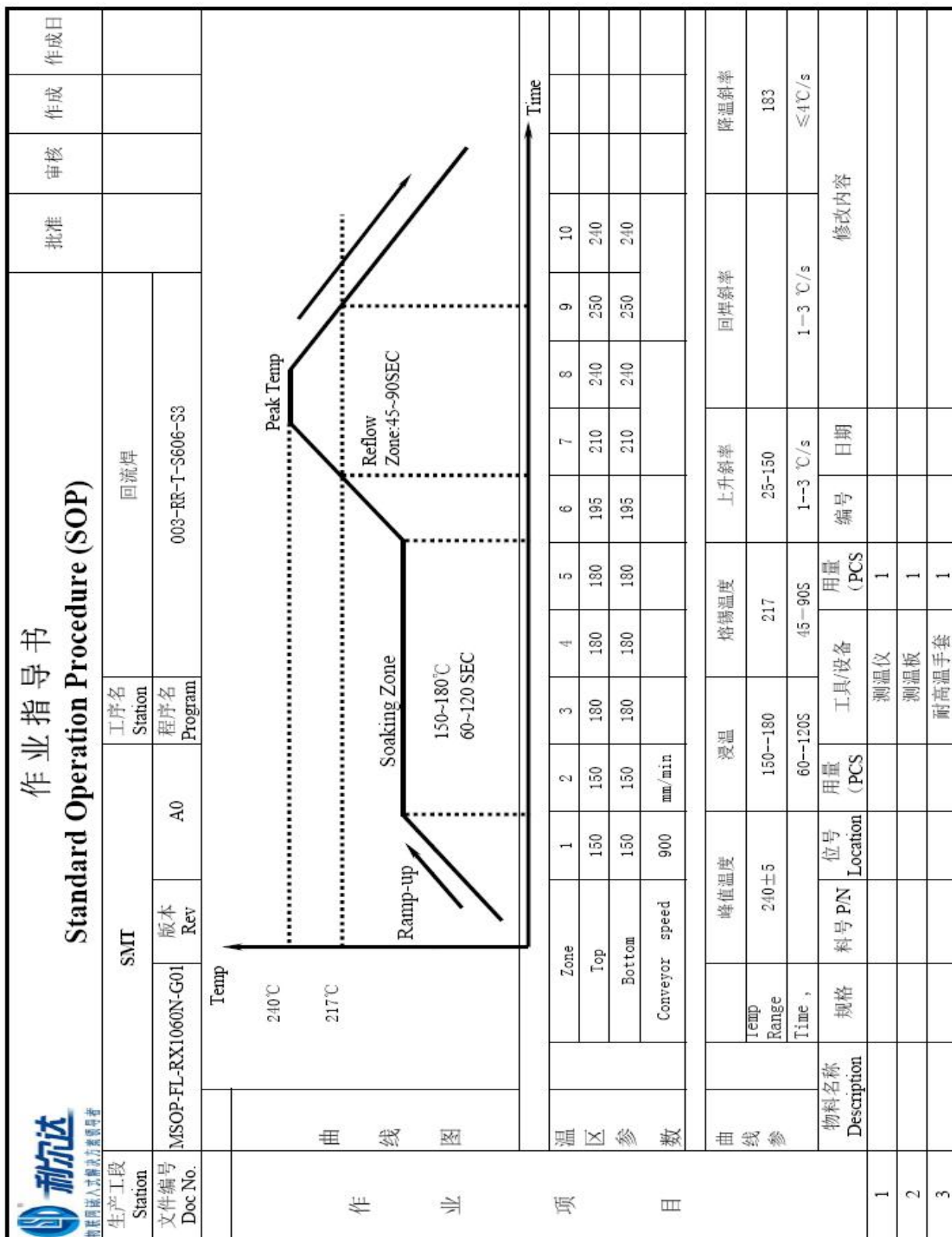


Figure 8.1 Lead-Free Reflow Soldering Operation Guide

8.1.3 Production process

During the production welding process or other processes that may directly contact the module, no organic solvents (such as alcohol, isopropanol, acetone, trichloroethylene, etc.) shall be used to wipe the module shielding cover; otherwise, it may cause the shielding cover to rust.

If spraying or potting is required for the module, please ensure that the spraying or potting materials used do not chemically react with the module's shield or PCB, and also ensure that the spraying or potting materials do not flow into the interior of the module.

8.1.4 Maintenance

If the module has defects such as cold solder joints or short circuits and requires repair, please follow the parameters below:

◆ Lead-free process: Soldering iron temperature $380 \pm 10^{\circ}\text{C}$, soldering iron contact time ≤ 5 seconds.

◆ Leaded process: Soldering iron temperature $350 \pm 10^{\circ}\text{C}$, soldering iron contact time ≤ 5 seconds.

It is not recommended to use a heat gun on the module to avoid affecting the module's performance.

8.2 Packaging specifications

The module's factory packaging uses a rubber wheel tape method, with the rubber wheel reference dimensions as follows:

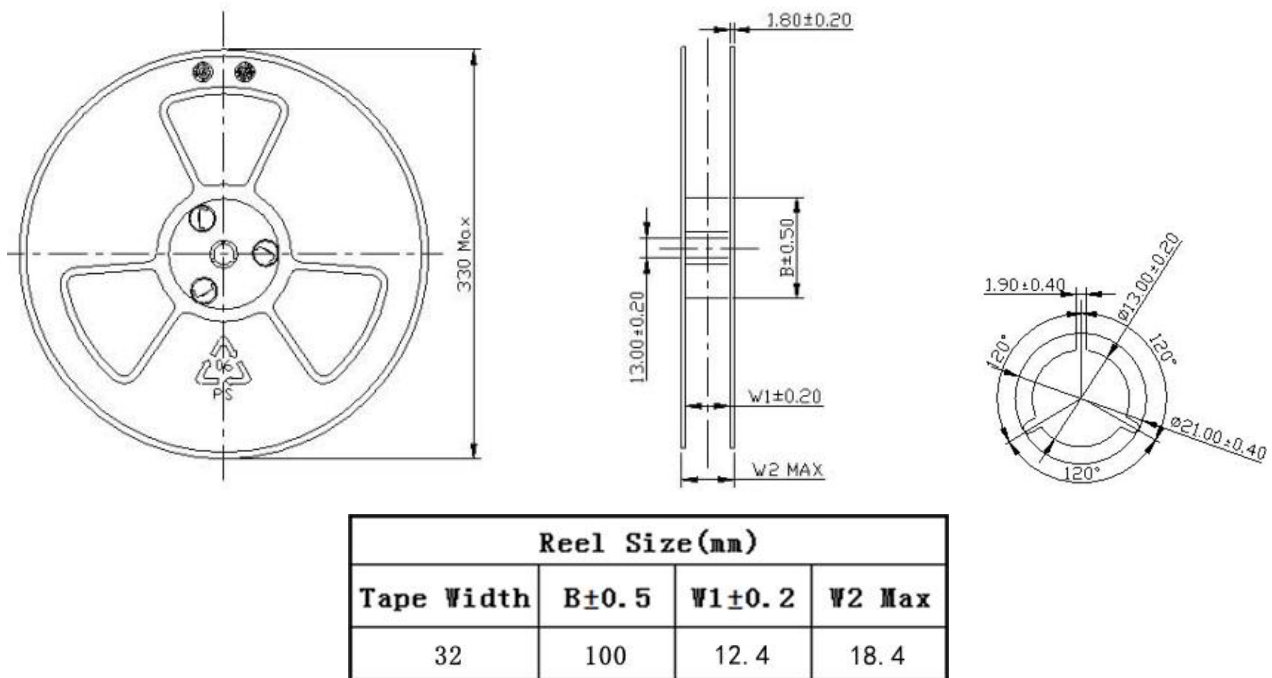
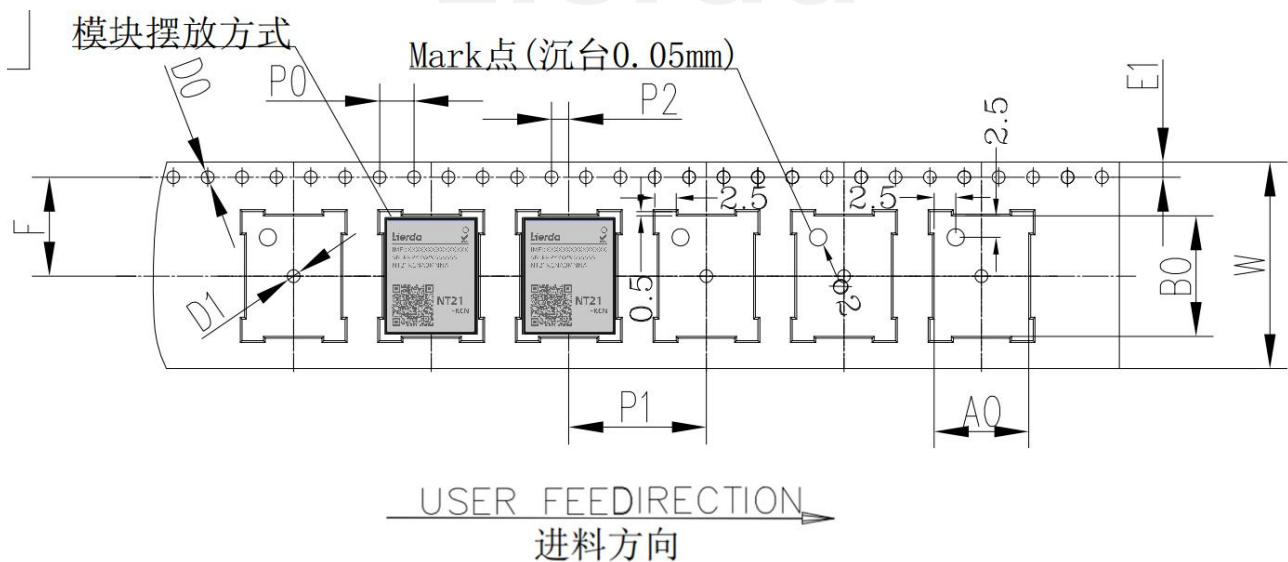


Figure 8.2 Carrier Belt Roller Dimension Diagram

The direction of tape feeding is as follows: (Refer to the diagram, the label content is subject to the actual product, and pay attention to the position of module PIN1).



ITEM	W	F	E1	D0	D1	P0	P2	T
DIM	24	11.5	1.75	φ1.5	φ1.5	4.00	2.00	0.30
TOLE	+0.30 -0.30	+0.10 -0.10	+0.10 -0.10	+0.10 -0.00	+0.20 -0.20	+0.10 -0.10	+0.10 -0.10	+0.05 -0.05

Figure 8.3 Carrier Tape Feeding Direction Indicator Diagram

8.3 Storage conditions

The module is shipped in the form of vacuum reel sealed bags, with a moisture sensitivity level of MSL 3. Storage conditions are as follows:

- ◆ When the temperature is below 40°C and humidity is below 90% (RH), solderability can be ensured for 12 months under well-sealed packaging conditions;
- ◆ After opening, ensure that the chip mounting is completed within 168 hours under the conditions of an ambient temperature below 30°C and relative humidity below 60% (RH);
- ◆ If the above conditions are not met, baking is required. The tape material should be baked at 60°C ± 5°C with humidity ≤ 60%RH for 48 hours; if accelerated baking is needed, the module must be taken out from the tape for separate baking (pay attention to ESD protection during removal), baked at 125°C ± 5°C with humidity ≤ 60%RH for 8 hours, with the total baking time less than 96 hours.

For more detailed guidance, please refer to the IPC/JEDEC J-STD-033 standard.

9 Related Documents and Terminology

Abbreviations

9.1 Relevant documents

The relevant document reference is as shown in the table below:

Table 9-1 Related Documents

Serial Number	Document Name	Annotation
[1]	IPC/JEDEC J-STD-033 Standard	

9.2 Terminology Abbreviation

The abbreviations and their related definitions involved in this document are as follows:

Table 9-2 Terminology Abbreviations

Abbreviation	Full English name	Full Chinese name
3GPP	3rd Generation Partnership Project	Third Generation Partner Program
ADC	Analog-to-Digital Converter	Analog-to-digital conversion
ANT	Antenna	Antenna
ASM	Antenna Switch Module	Antenna switch module
DAC	Digital -to- Analog Converter	Digital-to-analog conversion
DBG	Debug	Debugging
DC-DC	Direct Current - Direct Current	DC converter
DCXO	Digitally Controlled Crystal Oscillator	Digital Control Crystal Oscillator
DRX	Discontinuous Reception	Non-continuous reception
DTE	Data Terminal Equipment	Data terminal equipment
ECL	Equivalent Class Level	Network coverage level
ESD	Electro-Static discharge	Electrostatic discharge
EOS	Electrical Overtress	Electrical Overstress (Surge)
ESR	Equivalent Series Resistance	Equivalent series resistance
EVK	Evaluation Kit	Evaluation Toolkit
H-FDD	Half Frequency Division Duplexing	Frequency Division Half Duplex
FOTA	Firmware Over-The-Air	Remote firmware upgrade
GPIO	General-purpose input/output	General Input and Output
I/O	Input/Output	Input and output interface
I _{max}	Maximum Load Current	Maximum current

Inorm	Normal Current	Normal (typical) current
bps	Bits Per Second	Rate unit
LCC	Leadless Chip Carriers	Leadless chip carrier packaging
LCM	LCD Module	LCD display module
LDO	Low Dropout Regulator	Low dropout linear regulator
LGA	Land Grid Array	Grid array packaging
LwM2M	Lightweight Machine-To-Machine	Lightweight M2M Protocol
MCU	Mirco Controller Unit	Microcontroller Unit
MSL	Moisture Sentivity levels	Moisture Sensitivity Level
PCB	Printed Circuit Board	Printed Circuit Board
PCBA	Printed Circuit Board Assembly	Printed Circuit Board Assembly
PF	Paging Frame	Paging frame
PMU	Power Management Unit	Power Management Unit
PSM	Power Saving Mode	Energy-saving mode
RF	Radio Frequency	Radio Frequency
RoHS	Restriction of Hazardous Substances	Restriction of hazardous substances
RX	Receive	Received
TAU	Tracking Area Update	Tracking area update
TCP/IP	Transmission Control Protocol/Internet Protocol	Transmission Control Protocol/Internet Protocol
TVS	Transient Voltage Suppressor	Transient suppression diode
TX	Transmit	Send
UART	Universal Asynchronous Receiver & Transmitter	Universal Asynchronous Receiver and Transmitter (UART)

UDP/IP	User Datagram Protocol - Internet Protocol	User Datagram Protocol
URC	Unsolicited Result Code	Non-request result code
(U)SIM	(Universal) Subscriber Identification Module	(Generic) User Identity Recognition Module
VSWR	Voltage Standing Wave Ratio	Voltage Standing Wave Ratio
Vmax	Maximum Voltage Value	Maximum voltage
Vnorm	Normal Voltage Value	Normal (typical) voltage
Vmin	Minimum Voltage Value	Minimum voltage
VIHmax	Maximum Input High Level Voltage Value	Maximum input high level
VIHmin	Minimum Input High Level Voltage Value	Minimum input high level
VILmax	Maximum Input Low Level Voltage Value	Maximum input low level
VILmin	Minimum Input Low Level Voltage Value	Minimum input low level
VImax	Absolute Maximum Input Voltage Value	Maximum input level
VImin	Absolute Minimum Input Voltage Value	Minimum input level
VOHmax	Maximum Output High Level Voltage Value	Maximum output high level
VOHmin	Minimum Output High Level Voltage Value	Minimum output high level
VOLmax	Maximum Output Low Level Voltage Value	Maximum output low level
VOLmin	Minimum Output Low Level Voltage Value	Minimum output low level